K J Somaiya Institute of Technology
An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

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EXPERIMENT 5

Aim:

To implement Singly Linked List

Objective:

- To understand the concept and structure of a Singly Linked List in Data Structures and Algorithms (DSA).
- To learn how nodes are created and linked dynamically using pointers.
- To implement basic operations such as insertion, deletion, traversal, and searching in a singly linked list.
- To understand how a singly linked list differs from an array in terms of memory allocation and data organization.
- To develop a program that demonstrates the working of a singly linked list using assembly or a suitable programming language.

Software Used:

- Turbo C / Turbo C++
- DOSBox

Theory :

A **Singly Linked Li**st is a linear data structure in which elements, called nodes, are connected using pointers. Each node consists of two parts: a **data field**, which stores the actual value, and a **next pointer**, which stores the address of the next node in the list. The last node contains a **NULL** pointer, indicating the end of the list.

Unlike arrays, linked lists do not require contiguous memory locations. Memory is allocated dynamically whenever a new node is created, making linked lists useful when the size of the data structure changes frequently.

The first node of the list is accessed through a pointer called **HEAD**. The basic operations performed on a singly linked list include insertion, deletion, traversal, and searching. Insertion adds a new node to the list, deletion removes an existing node, traversal visits each node sequentially, and searching finds a particular element.

A singly linked list allows efficient insertion and deletion when the required position or node is already known. However, it supports only forward traversal because each node stores the address of the next node and not the previous node.

Program :

```c
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#include <malloc.h>

typedef struct node
{
    int data;
    struct node *next;
} node;
node *createList();
node *Insert_beg(node *head, int x);
node *Delete_beg(node *head);
void PrintList(node *head);
```

```c
void main()
{
    int choice, insert_option, delete_option, x;
    node *head = NULL;
    printf("Welcome to the implementation of the singly linked list ! \n");
    do
    {
        printf("Please select an operation to perform from the below list \n
        printf(" 1. Create a List \n 2. Insert a node \n 3. Delete a node \n
        printf("Enter your choice: ");
        scanf("%d", &choice);
        printf("\n \n");
        switch (choice)
```

```c
        case 1:
            head = createList();
            break;
        case 2:
            do
            {
    printf(" 1 Beginning of the List \n 2. Exit the insert operation \n");
                printf("Enter your choice: ");
                scanf("%d", &insert_option);
                switch (insert_option)
                {
                case 1:
                    printf("Enter the data to be inserted: ");
```

```
                scanf("%d", &x);
                head = Insert_beg(head, x);
                break;

            case 2:
                printf("Insert operation Exit");
                break;
            default:
                printf("Please enter a valid choide: 1, 2, 3, 4");
            }
        } while (insert_option != 2);
        printf("\n \n");
        break;
```

```
    case 3:
        do
        {
printf("Select a position from where you to want to delete the element \n");
printf(" 1. Beginning of the List \n 2. Exit the delete operation \n");
printf("Enter your choice: ");
scanf("%d", &delete_option);
            switch (delete_option)
            {
            case 1:
                head = Delete_beg(head);
                break;
```

```
            case 2:
                printf("Delete Operation Exit");
                break;
            default:
                printf("Please enter a valid choide: ");
            }
        } while (delete_option != 2);
        printf("\n \n");
        break;
    case 4:
        PrintList(head);
        break;
    case 5:
```

```
            printf("Exit: Program Finished !!");
            break;
        default:
            printf("Please enter a valid choide: ");
        }
    } while (choice != 5);
}

node *createList()
{
    node *head, *p;
    int i, n;
    head = NULL;
```

```
    printf("Enter the number of nodes: ");
    scanf("%d", &n);
    printf("Enter the data: ");
    for (i = 0; i <= n - 1; i++)
    {
        if (head == NULL)
        {
            p = head = (node *)malloc(sizeof(node));
        }
        else
        {
            p->next = (node *)malloc(sizeof(node));
            p = p->next;
```

```
        p->next = NULL;
        scanf("%d", &(p->data));
    }
    printf("\n \n");
    return (head);
}

node *Insert_beg(node *head, int x)
{
    node *p;
    p = (node *)malloc(sizeof(node));
    p->data = x;
    p->next = head;
```

```
    return (head);
}

node *Delete_beg(node *head)
{
    node *p, *q;
    if (head == NULL)
    {
        printf("Empty Linked List");
        return (head);
    }
    p = head;
    head = head->next;
```

```
}

void PrintList(node *head)
{
    node *p;
    printf("[ ");
    for (p = head; p != NULL; p = p->next)
    {
        printf("%d \t", p->data);
    }
    printf(" ]");
    printf("\n \n");
}
```

Output :

```
Welcome to the implementation of the singly linked list !
Please select an operation to perform from the below list
 1. Create a List
 2. Insert a node
 3. Delete a node
 4. Print the existing list
 5. Exit
Enter your choice: 1


Enter the number of nodes: 3
Enter the data: 10
20
30
```

```
Enter your choice: 1


Enter the number of nodes: 3
Enter the data: 10 20 30

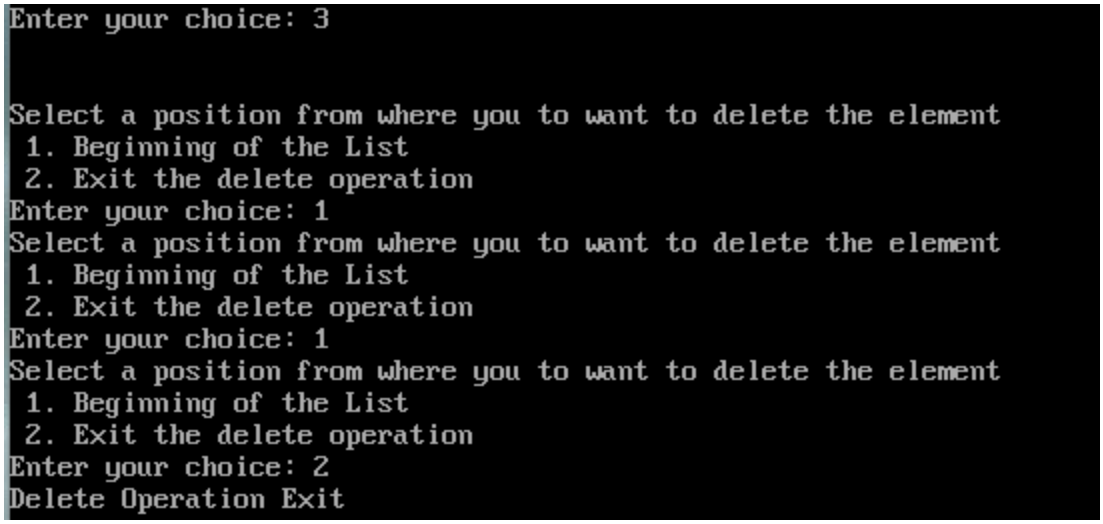

Please select an operation to perform from the below list
 1. Create a List
 2. Insert a node
 3. Delete a node
 4. Print the existing list
 5. Exit
Enter your choice: 4

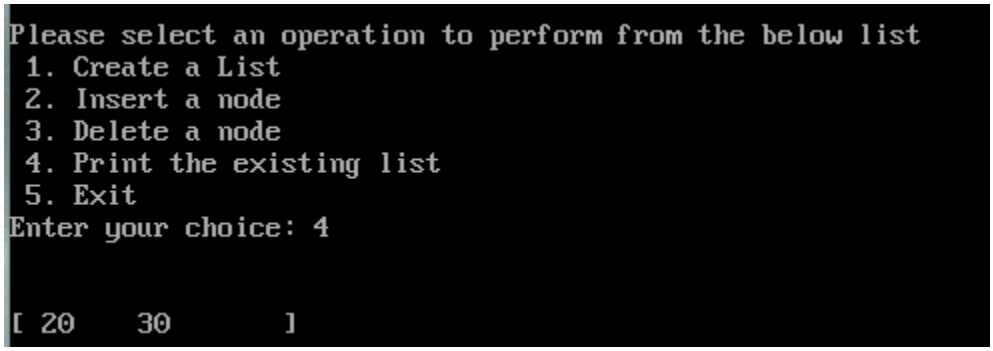

[ 10    20      30       ]
```

```
Enter your choice: 1
Enter the data to be inserted: 5
 1. Beginning of the List
 2. Exit the insert operation
Enter your choice: 2
Insert operation Exit

Please select an operation to perform from the below list
 1. Create a List
 2. Insert a node
 3. Delete a node
 4. Print the existing list
 5. Exit
Enter your choice: 4


[ 5     10      20      30       ]
```





Outcome :

The singly linked list data structure was studied and implemented successfully. The structure of nodes, the use of pointers to connect nodes, and the basic operations of insertion, deletion, and traversal were understood. The practical demonstrated how linked lists store and manage data dynamically.

Conclusion :

Thus, the implementation of a Singly Linked List was studied successfully. The practical helped in understanding dynamic memory organization, pointer-based connections between nodes, and fundamental linked-list operations. It also provided a clear understanding of how linked lists can be used to manage data efficiently in Data Structures and Algorithms.
